Module 11 Group Activity

Matrix Theory

1. (variation on NS 6.3.5) Let $A \in \mathbb{R}^{3\times 3}$ be such that there is no invertible $P \in \mathbb{R}^{3\times 3}$ such that $P^{-1}AP$ is upper triangular.

   (a) Explain why A cannot have three real eigenvalues.

   (b) Explain why this implies A must have exactly one real and two non-real eigenvalues.

   (c) Prove that there exists an invertible matrix $Q \in \mathbb{C}^{3\times 3}$ such that $Q^{-1}AQ$ is diagonal.

